Higher National Diploma in Engineering (Electrical)

4th Year 1st Semester Examination – 2023

EE 3215 – Broadband Communication

Instructions to Candidates:
Answer only five (05) questions.
Each question carries equal (20) marks
hours

No. of questions : 06
No. of pages : 03
Time : **03**

Q1.1.Define broadband communication and baseband communication? (4 marks)

2. Differentiate between the ADSL and VDSL (6 marks)

3. Fill the table given below (6 marks)

DSL types	Data rate down stream	Data rate up stream	Applications
ADSL			
HDSL			
VDSL			

4. Explain the following term (4 marks)

a.Baud rate

b.Bit rate

[Total 20 marks]

Q2.1.Draw the ADSL Architecture and explain the DSLAM, splitter and modem? (8 marks)

2. Briefly explain following (4 marks)

a.signal to noise ration

b. Quadrature Amplitude modulation (QAM)

3. What are the two types of modulation techniques are used in xDSL Technologies and explain any one of the modulation technique (6 marks)

4. What is multiplexing (2 marks)

[Total 20 marks]

Q3.1. what are the reasons to get low speed broadband? (4 marks)

2. What is the difference between ISDN and DSL? (4 marks)

3. Explain the broadband ISDN (6 marks)

4. ISDN provides a fully integrated digital service to users. These services fall into 3 categories what are they? Briefly explain these three categories of services.

(6 marks)

[Total 20 marks]

Q4.1.Briefly explain the following technologies

a.WiMAX (12 marks)

b.HSPA

c.Wifi

2. Distinguish between cellular phone and cordless phone (3 marks)

3. What is the principle of OFDM modulation scheme (3 marks)

4. State two advantages of WLAN (2 marks)

[Total 20 marks]

Q5.1.Draw the architecture of BPL and explain the function of BPL modem.

2. What are the advantages of BPL over other connection

3. Why uplink frequency is frequency is different from downlink frequency in Satelliate. Explain it

4. Discuss the evaluation of Mobile Network

Q6.1.What are the types of ATM services?

2. State the advantages and disadvantages of ATM Layer?

3. Explain the function of ATM switch?

4. Briefly explain the time division duplex and frequency division duplex

5. State some advantages of LTE over ADSL